



XMBA 200S: Data and Decisions

Chapter 19: Linear Patterns

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Course overview

Part 4: Building regression models

- Block 3 and 4
- You want to build a model to forecast outcomes from data you have
- What's the best **model you can build**?
- Today: constructing and interpreting OLS regression line
 - Future classes: inference; multivariate models

Outline

- Constructing the OLS regression line
- Interpreting and using the OLS model
- The importance of the residuals

Diamonds



You have a sample of several hundred emerald-cut, slightly “included” diamonds. Based on this evidence:

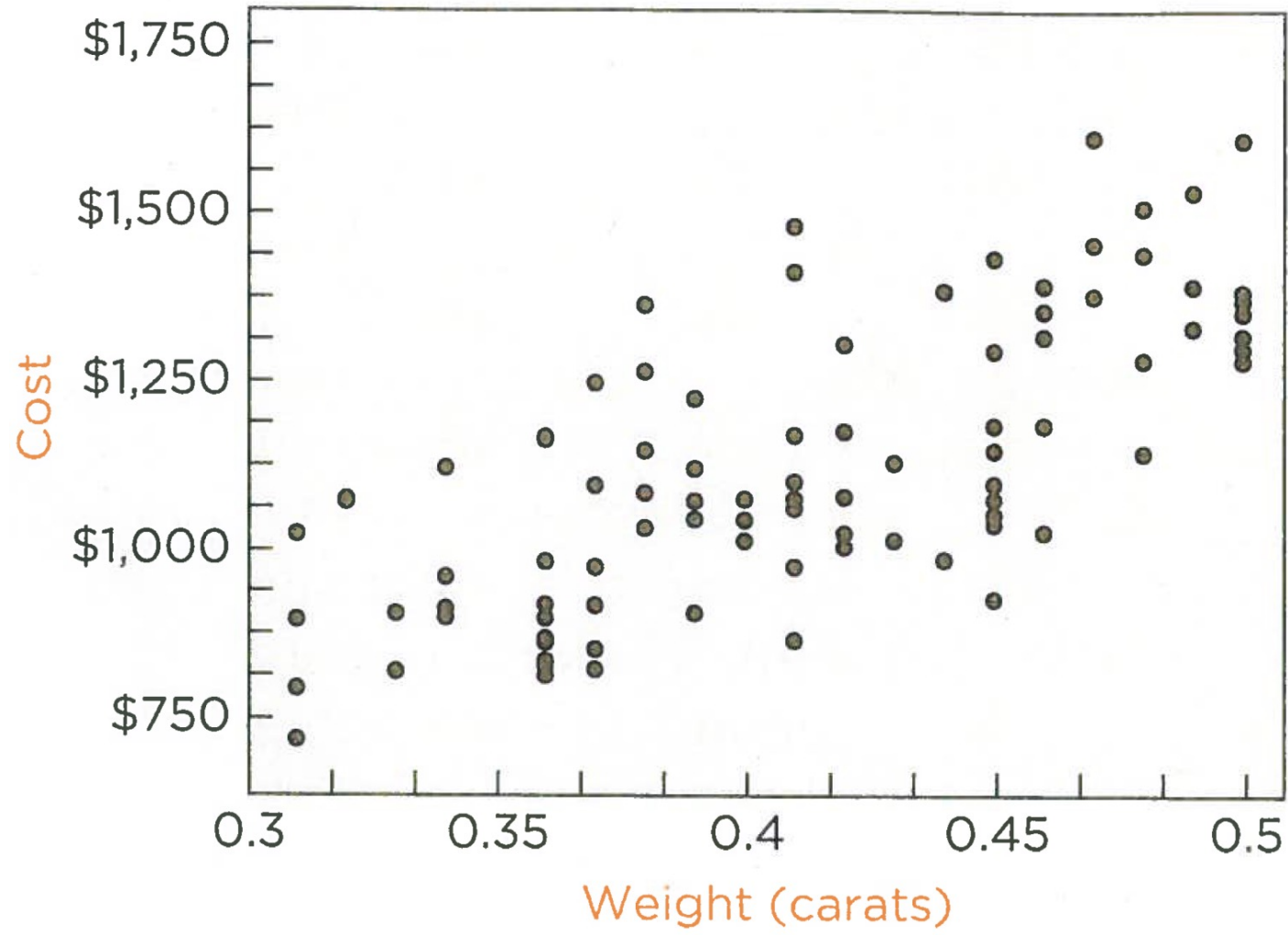
- What is the relationship between price and weight?
- What is the average price of diamonds that weigh 0.4 carat?
- How much more do diamonds that weigh 0.5 carat cost?

Fitting a line to data

Diamonds example

- To answer those questions, you want to build a linear model that links the weight of a diamond to price
 - Does a linear model make sense?
 - How do you build the model?

Scatterplot



What makes a good model?

- Many different lines can be drawn through cloud of data
- You want to estimate a relationship between weight and price that predicts price in the following way:
 1. The prediction isn't systematically wrong in one direction
 2. Distance between prediction and actual price is small

The Ordinary Least Squares (OLS) regression

Any linear model will look like the following:

$$\textit{Estimated Price} = b_0 + b_1 \textit{Weight}$$

Why? Because this is just the equation for a line . . .

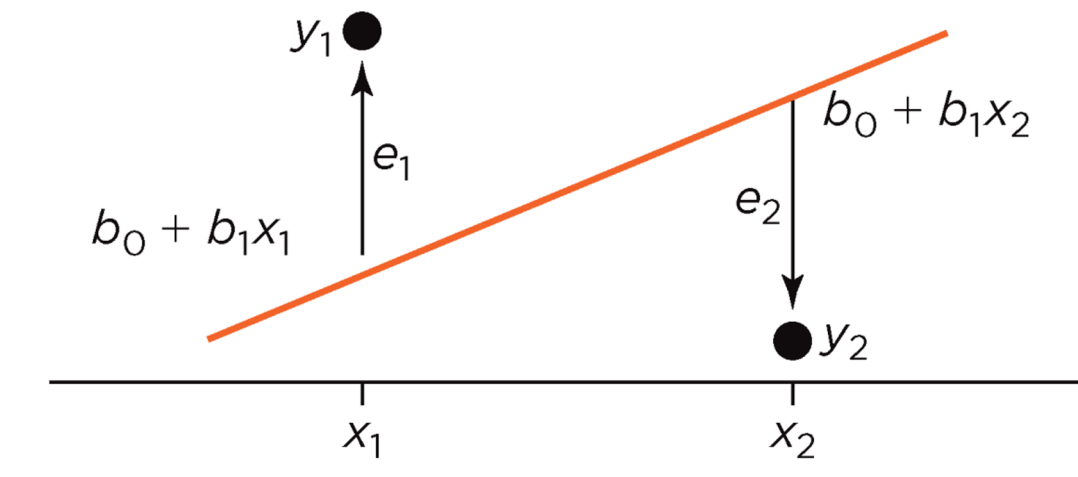
$$\hat{y} = b_0 + b_1 x$$

The question is how to choose b_0 and b_1 to make the model work well

Fitting a Line to Data

Residuals

Residuals are the vertical deviations from the data points to the fitted line, $e = y - \hat{y}$.



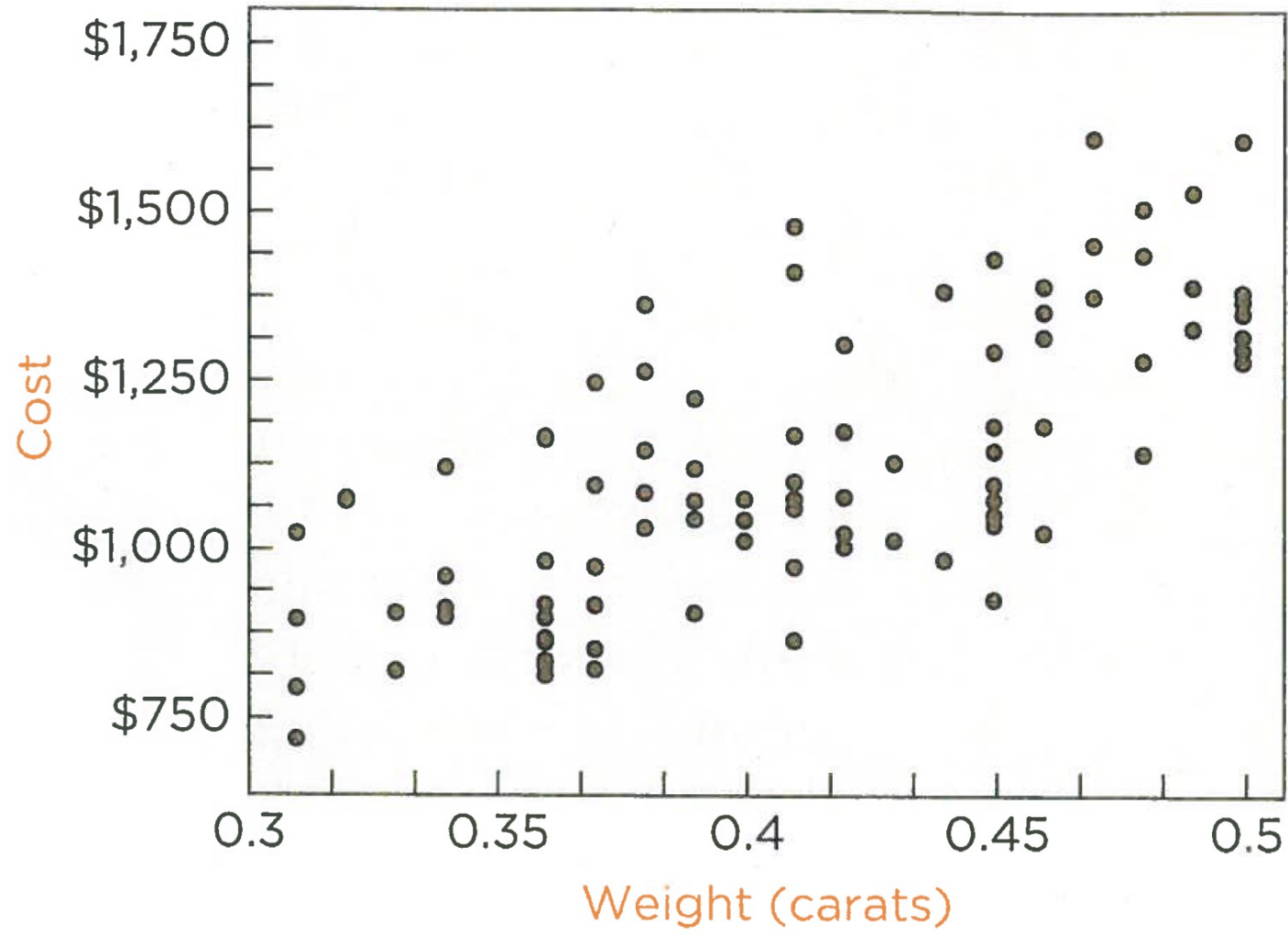
Fitting a Line to Data

Least Squares Regression

The regression line minimizes the sum of squared residuals.

More precisely, the intercept b_0 and slope b_1 are calculated to **minimize the squared residuals (“least squares”)**?

Scatterplot



Fitting a Line to Data

Ordinary Least Squares Regression

$$b_1 = r \frac{s_y}{s_x}$$

$$b_0 = \bar{y} - b_1 \bar{x}$$

Diamonds

Summary Statistics

Price (\$)			Weight (carats)	
Mean	1119.6		Mean	0.4094
Median	1080		Median	0.41
Std. Deviation	205.5273		Std. Deviation	0.05434
Kurtosis	-0.598		Kurtosis	-1.0039
Skewness	0.360		Skewness	0.0346
Minimum	714		Minimum	0.31
Maximum	1609		Maximum	0.5
Count	93		Count	93

Correlation Matrix

	Price (\$)	Weight (carats)
Price (\$)	1	
Weight (carats)	0.7131	1

Calculating the Fitted Line

Diamonds

$$b_1 = r \frac{s_y}{s_x} = 0.7131 \left(\frac{205.5273}{0.05434} \right) = 2697$$

$$b_0 = \bar{y} - b_1 \bar{x} = 1119.6 - (2697)(0.4094) = 15$$

Thus, the least squares regression equation for relating diamond prices to weight is,

$$\textit{Estimated Price} = 15 + 2697 \textit{ Weight}$$

Outline

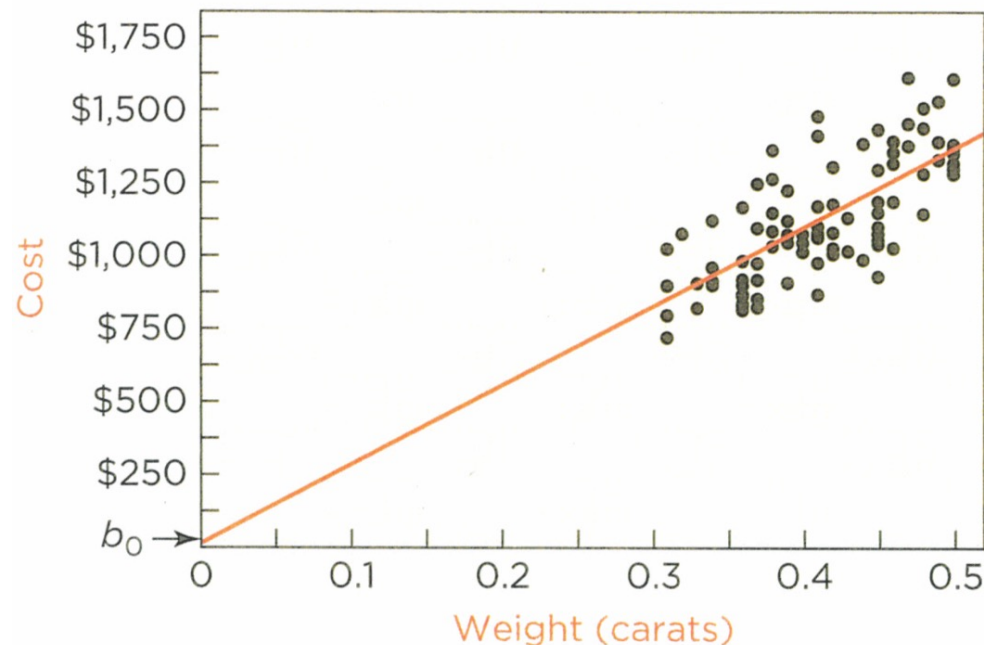
- Constructing the OLS regression line
- Interpreting and using the OLS model
- The importance of the residuals

Interpreting the Fitted Line

- Interpreting the y-Intercept b_0
 - The intercept is the portion of y that is present for all values of x
 - i.e., fixed cost, \$15, per diamond.
 - The intercept estimates the average response when $x = 0$
 - i.e. where the line crosses the y axis

Interpreting the Fitted Line

Interpreting the Intercept



In our sample we do not have any diamonds with near zero weight. Thus b_0 is an **extrapolation**, and should be interpreted with caution.

What does the OLS line tell you?

Interpreting the slope coefficient

The slope is the predicted change in the outcome when the explanatory variable changes by 1 unit (“change in y over change in x ”)

- In this case, a one carat change in diamond weight is associated with an increase in price by \$2,697

To find the effects of changes other than 1 unit, just multiply b_1 by the proper fraction/multiple

- In this case, a .2 carat change in diamond weight is associated with an increase in price by $0.2 \times 2697 = \$539$

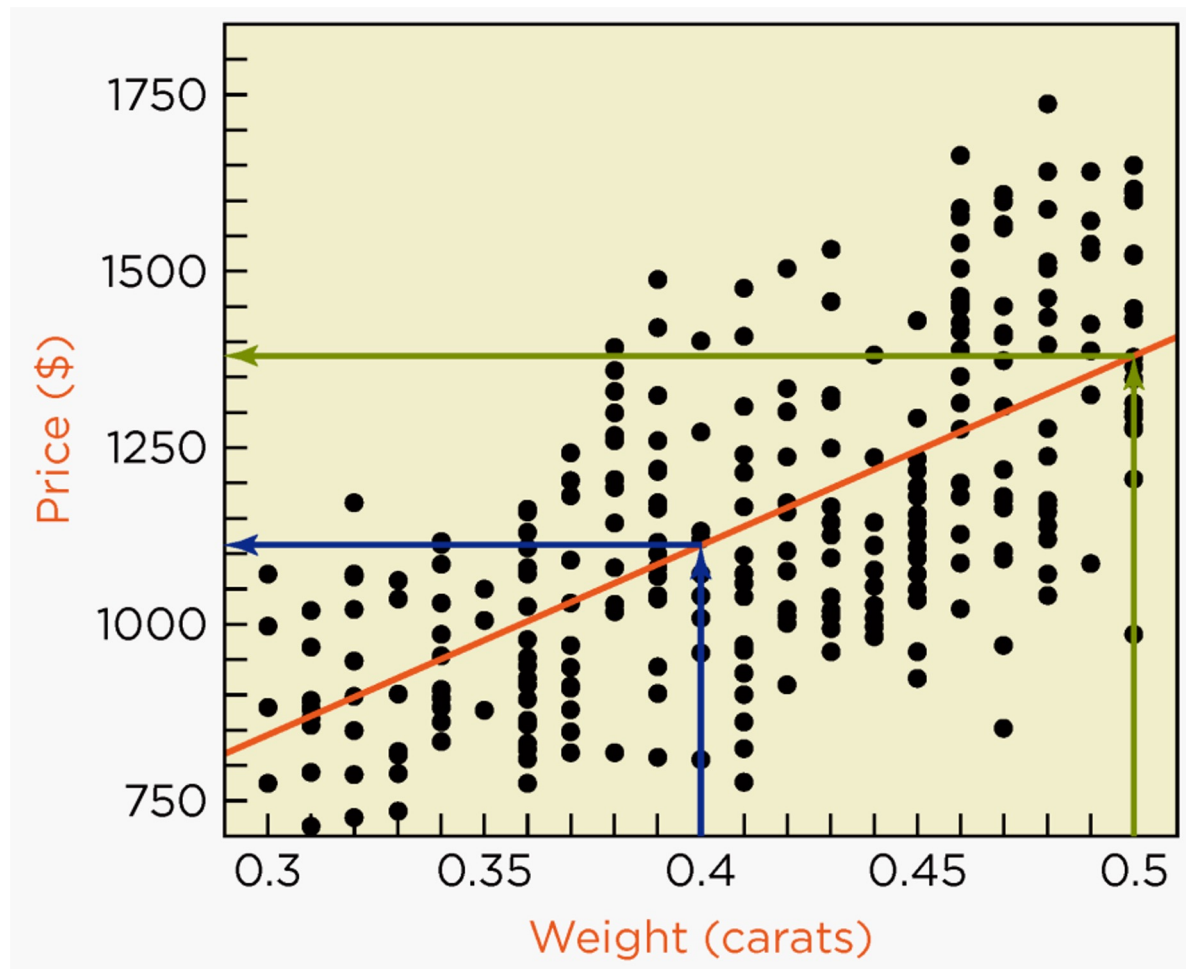
Interpreting the Fitted Line

Interpreting the Slope

- The slope estimates, in this example, the marginal cost (i.e., marginal cost is \$2,697 per carat).
- While tempting, it is not correct to describe the slope as *the change in y caused by changing x*.
 - This is an association (not causal)

Interpreting the Fitted Line

Diamond Example



Interpreting the Fitted Line

Prediction

What is the predicted price of diamonds that weigh 0.4 carat?

$$\begin{aligned}\text{Estimated Price} &= 15 + 2697 \text{ Weight} \\ \text{Estimated Price} &= 15 + 2697 (0.4) = \$1,094\end{aligned}$$

How much higher is the price for a 0.5 carat diamond?

\$270 more, on average.

Outline

- Constructing the OLS regression line
- Interpreting and using the OLS model
- The importance of the residuals

Properties of Residuals

Residuals

- Show variation that remains in the data after accounting for the linear relationship defined by the fitted line.
- Should be plotted against x to check for patterns.

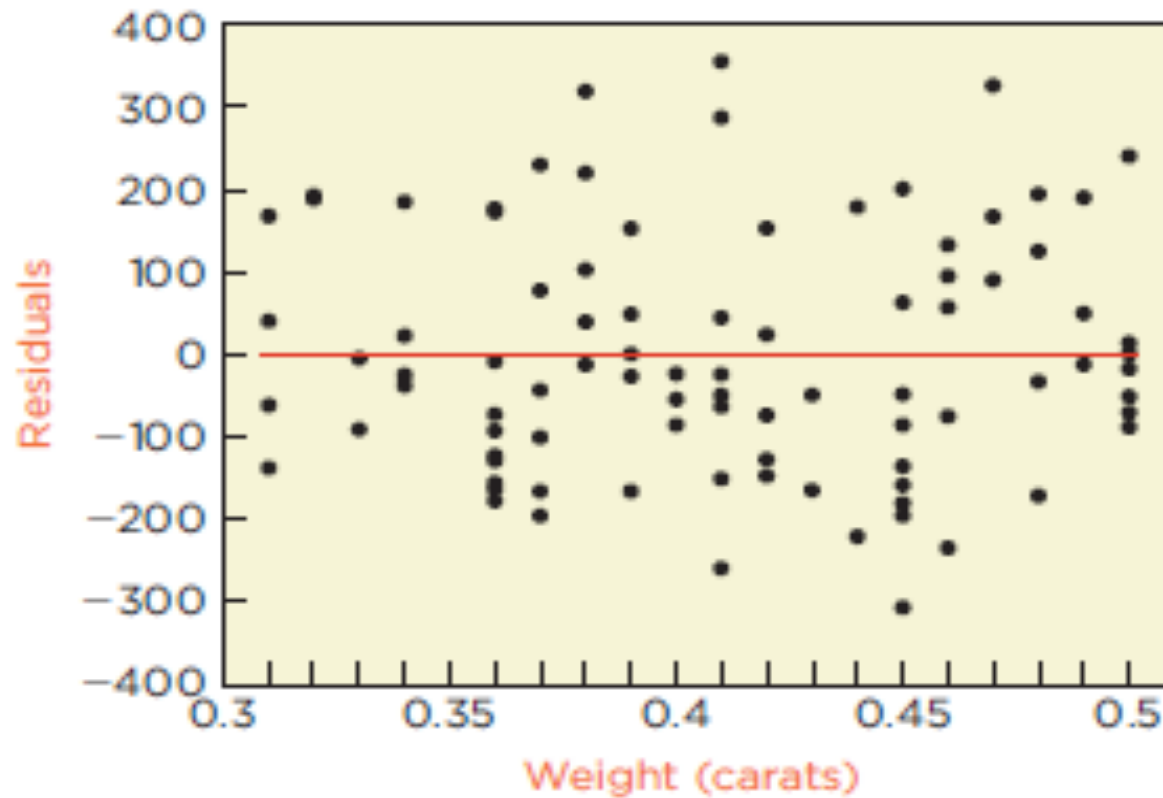
Is the linear model appropriate?

Checking residuals

- If the least squares model is appropriate then a plot of residuals against x should stretch out horizontally with “consistent vertical scatter”
- Can use a visual test for association to check for the absence of a pattern

Properties of Residuals

Residual Plot



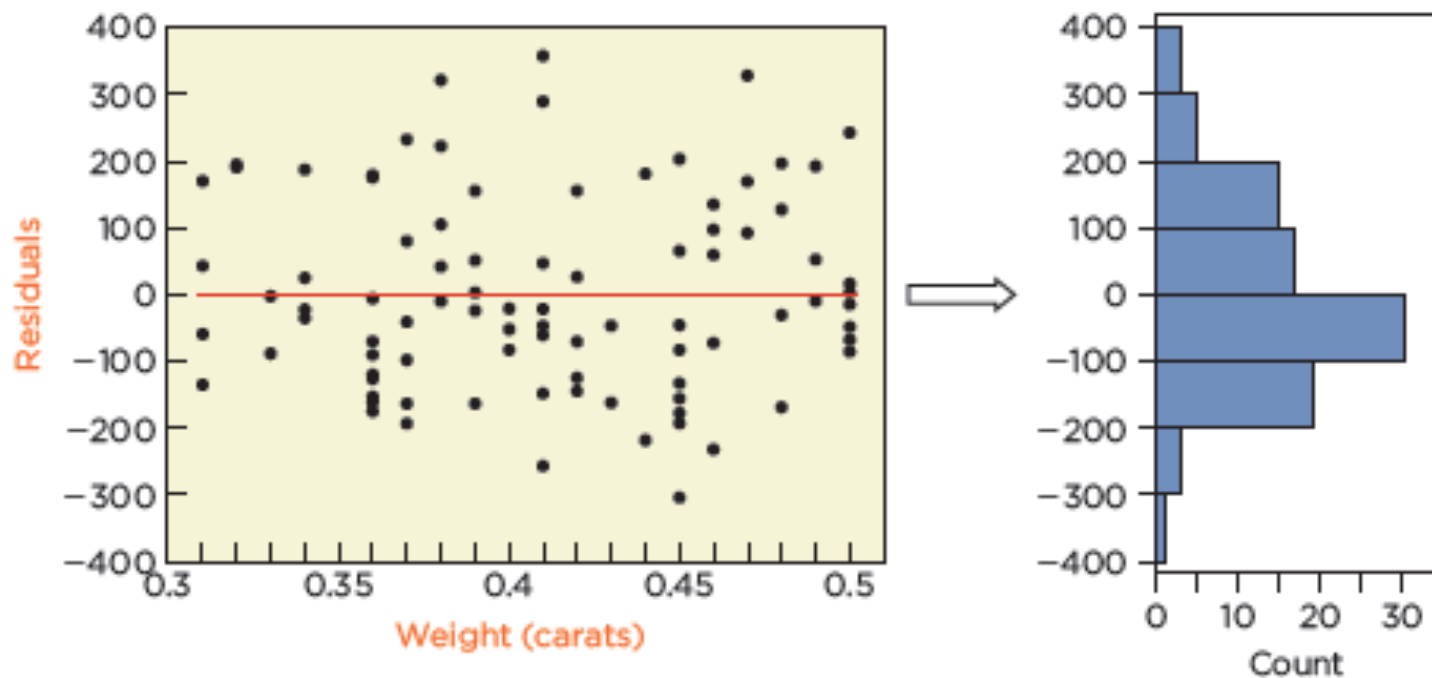
Properties of Residuals

Standard Deviation of Residuals (s_e)

- Measures how much residuals vary around fitted line.
- Excel calls this the standard error of the regression.
- For the diamond example, $s_e = 145$.

Properties of Residuals

Are the Residuals Approximately Normal?



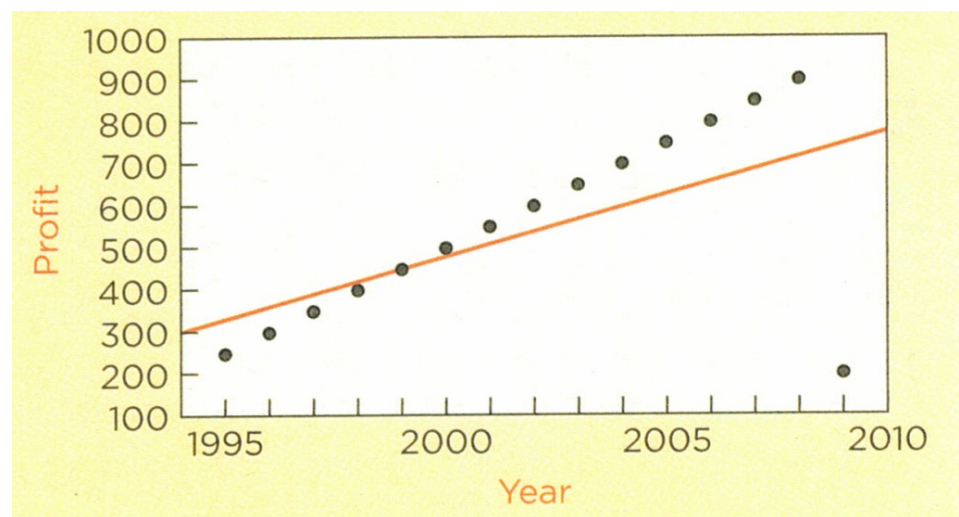
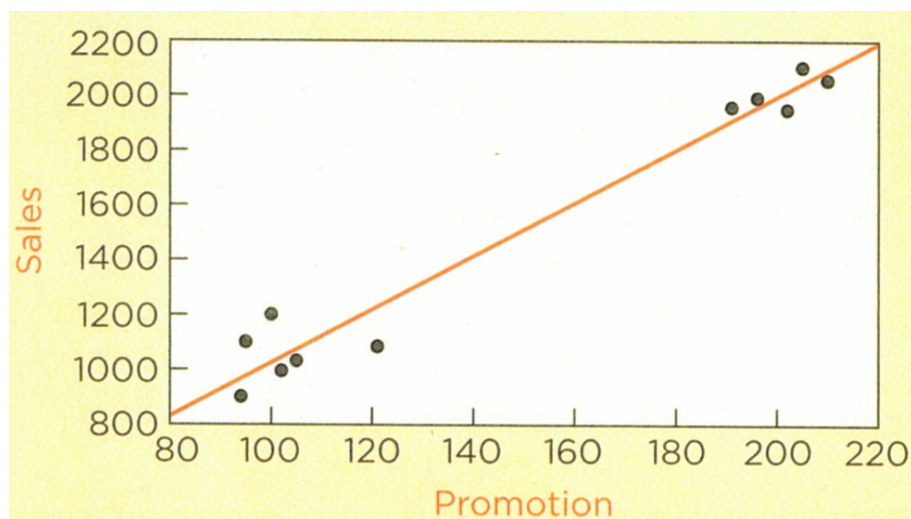
Explaining variation: R-squared (r^2)

- The residuals are what your model fails to explain
- What your model does explain is measured by the r^2 – the square of the correlation between x and y
 - The fraction of the variation in y accounted for by the least squares regression line
- For the diamond example, $r^2 = 0.509$
 - Thus, the fitted line explains 50.9% of the variation in price
- NOTE: r^2 alone does not tell you whether you have a good model
 - Causality?
 - Is the linear fit appropriate?

Explaining Variation

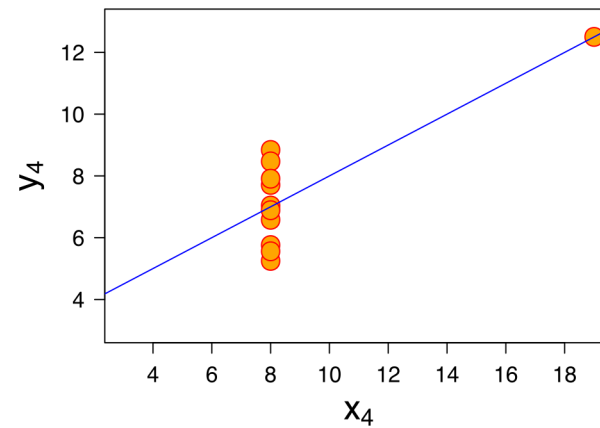
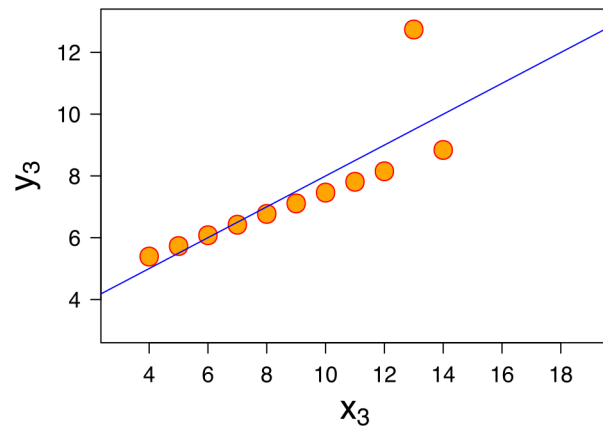
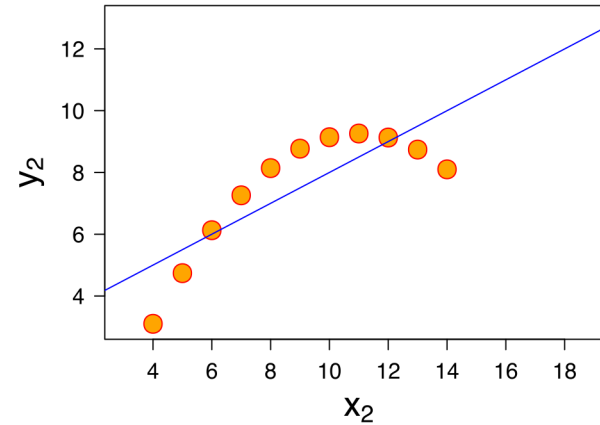
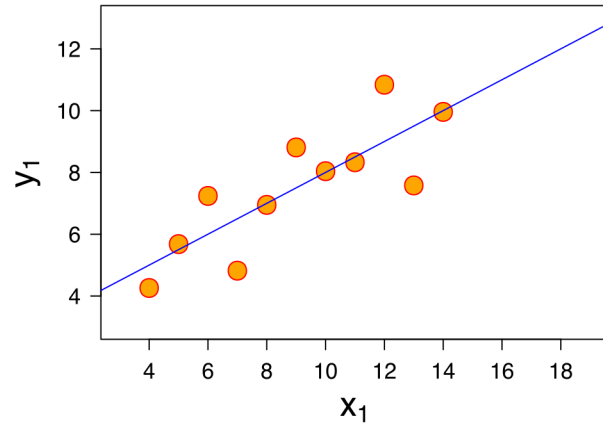
R-squared (r^2)

Don't trust summaries like r^2 without looking at plots.



Anscombe's Quartet

Plot Your Data!



Plot Your Data!

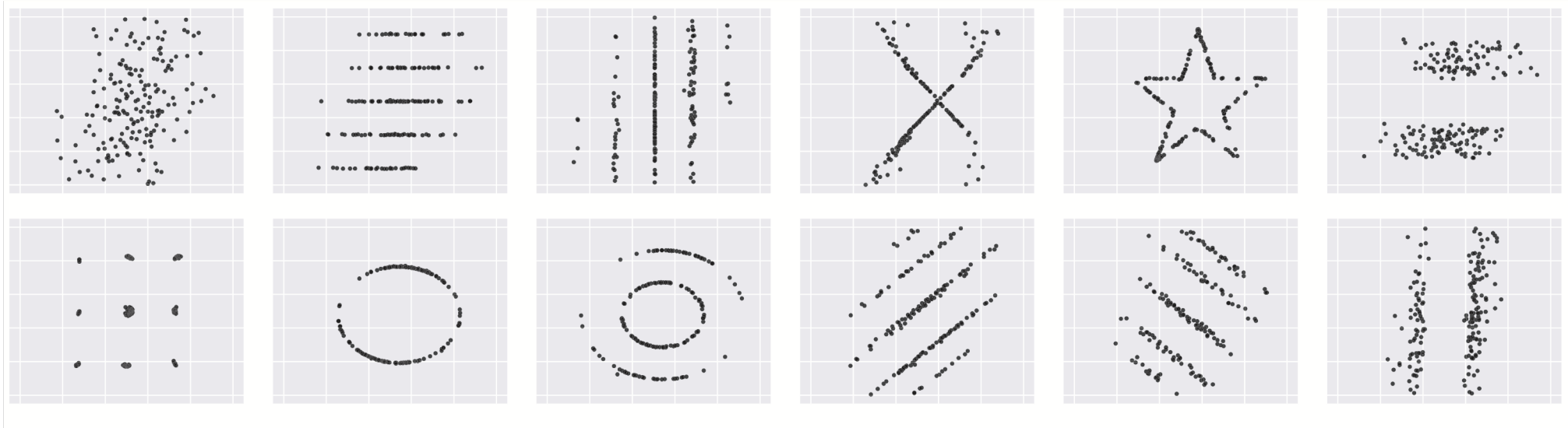


Figure 1. A collection of data sets produced by our technique. While different in appearance, each has the same summary statistics (mean, std. deviation, and Pearson's corr.) to 2 decimal places. ($\bar{x}=54.02$, $\bar{y}=48.09$, $sd_x=14.52$, $sd_y=24.79$, Pearson's $r=+0.32$)

Same Stats, Different Graphs: Generating Datasets with Varied Appearance and Identical Statistics through Simulated Annealing

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R-Squared is Not Causality

- We just showed that R-squared does not tell you whether a relationship is linear
- A high R-squared also does **not** tell you that a relationship is causal. A low R-squared does not mean that there is no causal relationship.
 - High R-squared, not causal: Nicolas Cage movies & drownings
 - Low R-squared, causal: smoking on life expectancy
- R-squared says how much of the variation in y is explained by x
 - Reminder: some things are harder to predict than others

Checklist for simple regression

- Always look at the scatterplot.
- Describe the intercept and slope using units of the data.
- Limit predictions to the range of observed conditions.
- Can you think of any lurking variable?
 - Use your intuition and understanding of context to evaluate other explanatory variables that might explain linear association between x and y
- Is the relationship linear?
 - Use a scatterplot to see if the pattern resembles a straight line
- Is the residual variation random?
 - Use the residual plot to make sure no pattern exists